

GOOGLE COLAB READY

Python for Data Science, ML & AI

Classes 01–03 combined: Python Basics → NumPy →
Pandas

**Beginner friendly • Big fonts • One script • No
external data**

What students will learn

A complete beginner path from Python fundamentals to clean data and outputs.

01

Python syntax

Variables, data types, conditions, loops, functions

02

NumPy thinking

Arrays, vectorized calculations and linear algebra

03

Pandas workflow

Create, inspect, clean and transform a dataset

04

Data outputs

Summaries, charts and CSV export for reproducible work

Teaching style: simple examples first, then practical health-style data analysis.

One-script roadmap

The whole lesson runs from top to bottom in one Python file.



```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Then: create data -> clean data -> analyse -> plot -> save
```

Big idea: every analysis script follows a clear pipeline.

Variables and data types

Teach students how Python stores information.

```
name = "Salek Data Lab"  
year = 2026  
pi = 3.14159  
is_ready = True  
  
print(type(name))
```

str

Text

int

Whole number

float

Decimal

bool

True / False

type()
checks class

Beginner message: variables are named boxes that hold data.

Core data structures

Four simple containers students will use constantly.

List

Ordered, changeable

```
scores = [72, 81, 90]
```

Dictionary

Key–value pairs

```
student = {"age": 22}
```

Tuple

Fixed ordered values

```
coordinates = (24.89, 91.87)
```

Set

Unique values

```
set(scores)
```

Use lists for columns, dictionaries for records, and DataFrames for complete datasets.

Logic, loops and functions

Move from single commands to reusable thinking.

```
for score in scores:
    if score >= 80:
        grade = "A"
    elif score >= 70:
        grade = "B"
    else:
        grade = "C"
```

```
def calculate_bmi(weight,
height):
    return weight / (height**2)

bmi = calculate_bmi(67, 1.72)
```

```
def safe_divide(a, b):
    try:
        return a / b
    except ZeroDivisionError:
        return "Cannot divide by
zero"
```

Key teaching point: conditions make decisions, loops repeat tasks, functions avoid repeated code.

NumPy makes calculations fast

Students learn arrays, summaries and vectorized operations.

```
x = np.array([1, 2, 3, 4, 5])  
  
np.mean(x)  
np.median(x)  
np.std(x, ddof=1)  
  
x**2
```



↓ vectorized operation



**Instead of writing many loops,
NumPy applies formulas to entire
arrays.**

Linear algebra in Python

A simple 2×2 example introduces matrix thinking.

$$\mathbf{A} \times \mathbf{x} = \mathbf{b}$$

```
A = np.array([[2, 1],  
              [1, 3]])  
  
b = np.array([5, 6])  
  
solution = np.linalg.solve(A,  
b)
```

solve()

Finds unknown x

det()

Measures matrix scale

inv()

Finds inverse matrix

Application idea: linear algebra is the mathematical language behind ML models.

Create a synthetic health dataset

A realistic practice dataset without downloading external files.

```
df = pd.DataFrame({  
    "id": np.arange(1, n + 1),  
    "age": rng.integers(18, 80, n),  
    "sex": rng.choice(..., n),  
    "bmi": rng.normal(25, 4, n),  
    "smoker": rng.choice(..., n),  
    "outcome": rng.binomial(1, 0.25, n)  
})
```

id

age

sex

bmi

smoker

outcome

500

rows

6

variables

Data cleaning workflow

Teach cleaning as a sequence, not as random commands.



```
df["sex"] = df["sex"].str.strip().str.title()
df = df.drop_duplicates(subset="id")
df["bmi"] = df["bmi"].fillna(df["bmi"].median())
```

From cleaned data to results

Students practice the most useful summary commands.

```
df["sex"].value_counts()

df.groupby("sex")["bmi"].mean()

pd.crosstab(df["sex"], df["outcome"])

pd.pivot_table(df, values="outcome",
index="age_group", columns="sex")
```

Frequency

Count categories

Groupby

Mean within groups

Crosstab

Two-way table

Pivot

Compact result
table

**Because outcome is 0/1,
mean(outcome) = prevalence.**

Visualize and export results

End every class with an output students can save.

```
pivot.plot(kind="bar", figsize=(8, 5))  
plt.title("Outcome Prevalence")  
plt.ylabel("Proportion")  
plt.show()  
  
df.to_csv("/content/cleaned_health_data.csv", index=False)
```

Bar chart



CSV exported

Mini assignment

Use this slide at the end of the session or as homework.

- 1 Calculate mean, minimum and maximum from a list
- 2 Create a participant dictionary with ID, age, sex and BMI
- 3 Classify ages using if / elif / else
- 4 Create a NumPy array and compute summary statistics
- 5 Use Pandas to summarise BMI and outcome prevalence
- 6 Export the final cleaned dataset as a CSV file

Next class idea: Exploratory Data Analysis + first machine-learning model

Class Completed

Python Basics • NumPy • Pandas • Cleaning • Visualization

Salek Data Lab